Power BI + Snowflake Visualization

Hospital Operations & Patient Analytics Dashboard

1. Project objective

The primary objective of this project is to analyze hospital operations and patient visit records using Snowflake and Microsoft Power BI.

The dashboard will help hospital management understand patient distribution, department performance, doctor workloads, treatment costs, admission trends and payment patterns.

By analyzing this information, hospital administrators can identify departments receiving the highest number of patients, evaluate resource utilization, monitor treatment expenses and make informed operational decisions.

Project description

A multi-specialty hospital wants to monitor its daily operations and understand how different departments are performing.

The hospital maintains information about patients, doctors, departments, appointments, treatments, admission types and billing amounts.

However, management needs a centralized dashboard to understand this information instead of manually analyzing individual records.

We will build an interactive dashboard that connects to a Snowflake database and uses Power BI for data transformation, calculations and visualization.

Visualizations:

No.	Analysis	What we want to identify
1	Patient distribution by city	Which city contributes the most patients?
2	Department-wise patient count	Which department receives the most patients?
3	Doctor-wise patient visits	Which doctors handle the most patients?
4	Monthly patient trends	How do patient visits change over time?
5	Admission type analysis	How many patients are OPD, IPD and Emergency?
6	Revenue by department	Which department generates the highest billing revenue?
7	Payment method analysis	Which payment method is most commonly used?
8	Treatment cost analysis	Which treatments have the highest average cost?

Dataset design

We'll create a single table named HOSPITAL_VISITS.

The dataset will contain 25 records with the following columns:

Column	Description
VISIT_ID	Unique visit identifier
VISIT_DATE	Date of hospital visit
PATIENT_ID	Unique patient identifier
PATIENT_NAME	Patient name
CITY	Patient's city
DEPARTMENT	Hospital department
DOCTOR_NAME	Treating doctor
ADMISSION_TYPE	OPD, IPD or Emergency
TREATMENT	Type of treatment
CONSULTATION_FEE	Doctor consultation charges
TREATMENT_CHARGE	Additional treatment charges
PAYMENT_METHOD	Cash, UPI, Card or Insurance
VISIT_STATUS	Completed, Admitted or Discharged

Snowflake database setup

Create database, schema and warehouse



Select them:

```
USE WAREHOUSE UDAY_HOSPITAL_WH;

USE DATABASE UDAY_HOSPITAL_DB;

USE SCHEMA HOSPITAL_SCHEMA;
```

Results (just now) — Table | Chart | Pivot — 1 row — 29ms

	status
1	Statement executed successfully.

Create the table

```
CREATE OR REPLACE TABLE HOSPITAL_VISITS (
    VISIT_ID VARCHAR(10),
    VISIT_DATE DATE,
    PATIENT_ID VARCHAR(10),
    PATIENT_NAME VARCHAR(50),
    CITY VARCHAR(30),
    DEPARTMENT VARCHAR(30),
    DOCTOR_NAME VARCHAR(50),
    ADMISSION_TYPE VARCHAR(20),
    TREATMENT VARCHAR(50),
    CONSULTATION_FEE NUMBER(10,2),
    TREATMENT_CHARGE NUMBER(10,2),
    PAYMENT_METHOD VARCHAR(20),
    VISIT_STATUS VARCHAR(20)
);
```

Results (just now) — Table | Chart | Pivot — 1 row — 168ms

	status
1	Table HOSPITAL_VISITS successfully created.

Insert the data :

```
('V018','2026-04-13','P018','Manoj','Mumbai','Dermatology','Dr. Kavya','OPD','Skin Treatment',1200,12500,'Card','Completed'),

('V019','2026-04-18','P019','Keerthi','Chennai','Neurology','Dr. Mehta','Emergency','MRI Scan',1800,45000,'Insurance','Admitted'),

('V020','2026-04-25','P020','Suresh','Bangalore','General Medicine','Dr. Arjun','OPD','Blood Test',1000,8000,'UPI','Completed'),

('V021','2026-05-03','P021','Neha','Kolkata','Cardiology','Dr. Sharma','IPD','ECG',1500,24000,'Insurance','Discharged'),

('V022','2026-05-08','P022','Ramesh','Pune','Pediatrics','Dr. Priya','OPD','General Checkup',1200,9500,'Cash','Completed'),

('V023','2026-05-14','P023','Swathi','Hyderabad','Orthopedics','Dr. Reddy','Emergency','Fracture Treatment',1800,21000,'Card','Admitted'),

('V024','2026-05-20','P024','Tarun','Delhi','General Medicine','Dr. Arjun','OPD','Blood Test',1000,17000,'UPI','Completed'),

('V025','2026-05-27','P025','Lavanya','Mumbai','Neurology','Dr. Mehta','IPD','CT Scan',1500,30000,'Insurance','Discharged');
```

Results (just now) — Table | Chart | Pivot — 1 row — 583ms

	number of rows inserted
1	25

Verifying

```
SELECT * FROM HOSPITAL_VISITS;
```

Results (just now) — Table | Chart | Pivot — 25 rows — 93ms

	VISIT_ID	VISIT_DATE	PATIENT_ID	PATIENT_NAME	CITY	DEPARTMENT	DOCTOR_NAME	ADMISSION_TYPE	TREATMENT
	V0... 4.0% V0... 4.0% +23 more	1/5/2... 5/27/...	P001 4.0% P002 4.0% +23 more	Ajay 4.0% Akhil 4.0% +23 more	Hyde... 20.0% Delhi 16.0% +5 more	Cardiology 20.0% Neurology 20.0% +4 more	Dr. Mehta 20.0% Dr. Sharma 20.0% +4 more	OPD 48.0% IPD 28.0% +1 more	Blood T... 16.0% General... 16.0% +9 more
1	V001	2026-01-05	P001	Rahul	Hyderabad	Cardiology	Dr. Sharma	OPD	ECG
2	V002	2026-01-08	P002	Priya	Chennai	Orthopedics	Dr. Reddy	IPD	Fracture Treatment

Feedback

Patient count by city

```sql
SELECT
    CITY,
    COUNT(*) AS TOTAL_PATIENTS
FROM HOSPITAL_VISITS
GROUP BY CITY
ORDER BY TOTAL_PATIENTS DESC;
```

Results (just now)

Table | Chart | Pivot — 7 rows — 22ms

	CITY	TOTAL_PATIENTS
1	Hyderabad	5
2	Mumbai	4
3	Delhi	4
4	Chennai	3

Department-wise patient count

```sql
SELECT
    DEPARTMENT,
    COUNT(*) AS PATIENT_COUNT
FROM HOSPITAL_VISITS
GROUP BY DEPARTMENT
ORDER BY PATIENT_COUNT DESC;
```

Results (just now)

Table | Chart | Pivot — 6 rows — 23ms

	DEPARTMENT	PATIENT_COUNT
1	Cardiology	5
2	Neurology	5
3	Orthopedics	4
4	Pediatrics	4
5	General Medicine	4

Feedback

Doctor-wise visits

```sql
SELECT
    DOCTOR_NAME,
    COUNT(*) AS TOTAL_VISITS
FROM HOSPITAL_VISITS
GROUP BY DOCTOR_NAME
ORDER BY TOTAL_VISITS DESC;
```

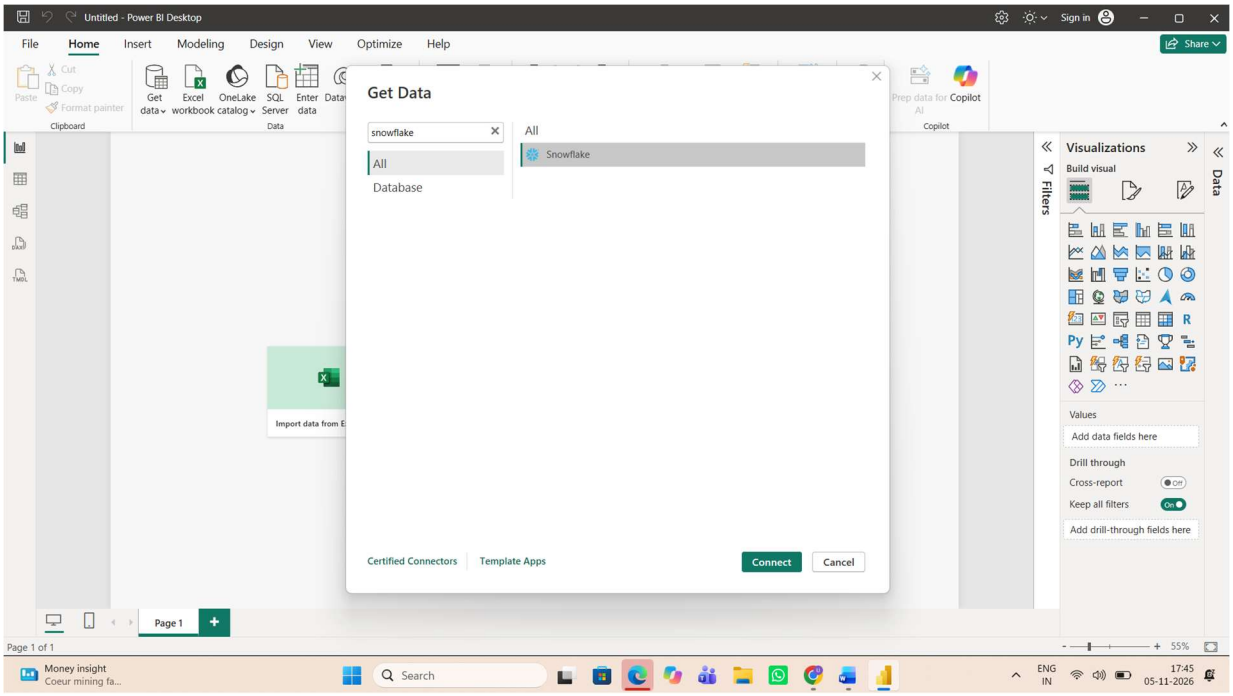
Results (just now)

Table | Chart | Pivot — 6 rows — 269ms

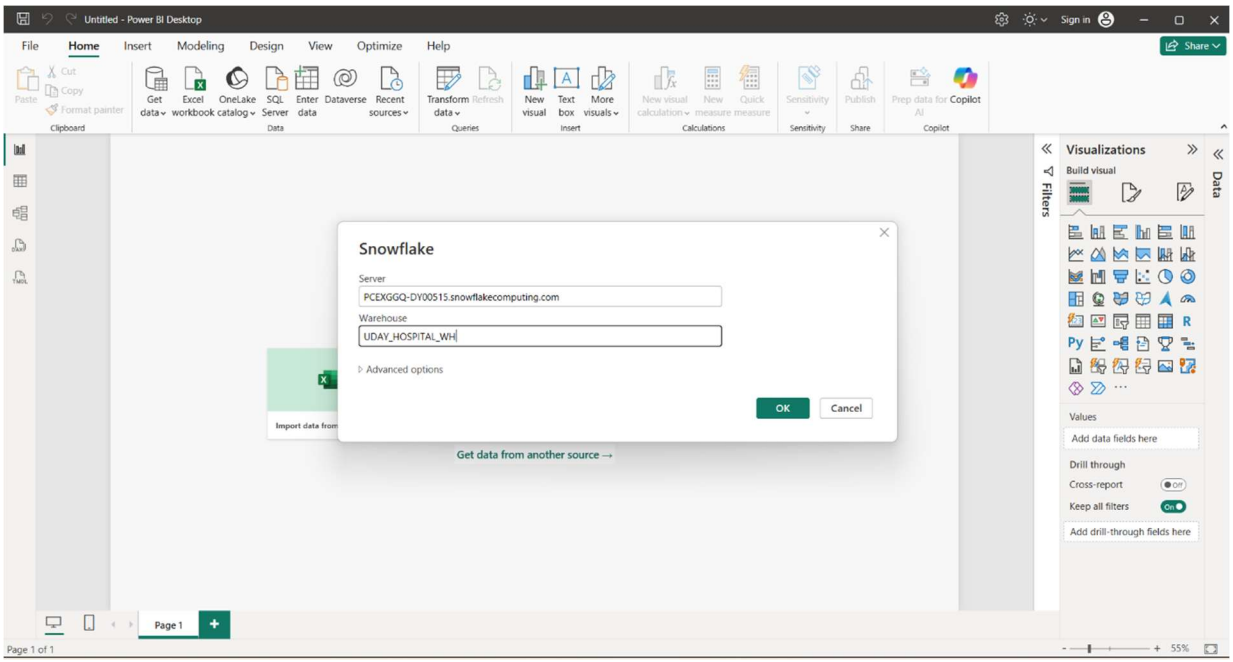
	DOCTOR_NAME	TOTAL_VISITS
1	Dr. Sharma	5
2	Dr. Mehta	5
3	Dr. Reddy	4
4	Dr. Priya	4
5	Dr. Arjun	4

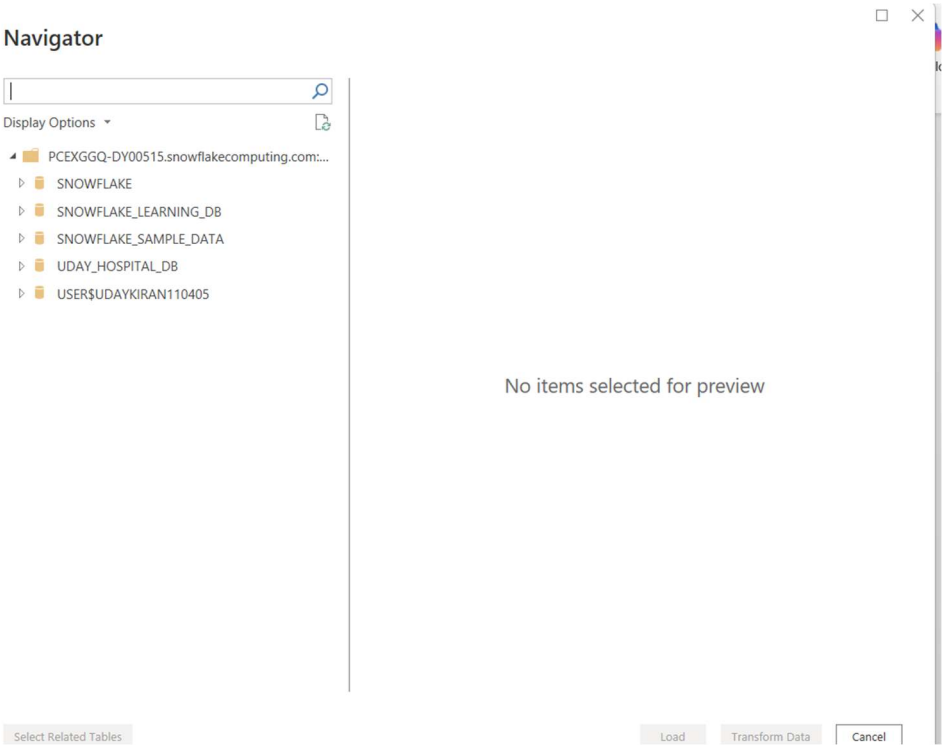
Feedback

Connecting the snowflake to the powerBI:

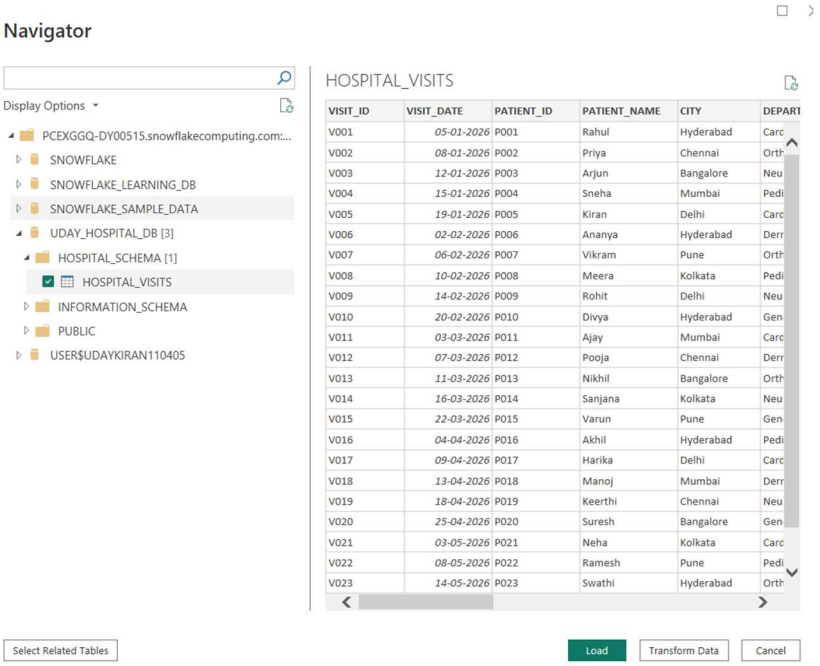


Provide the Snowflake Data :

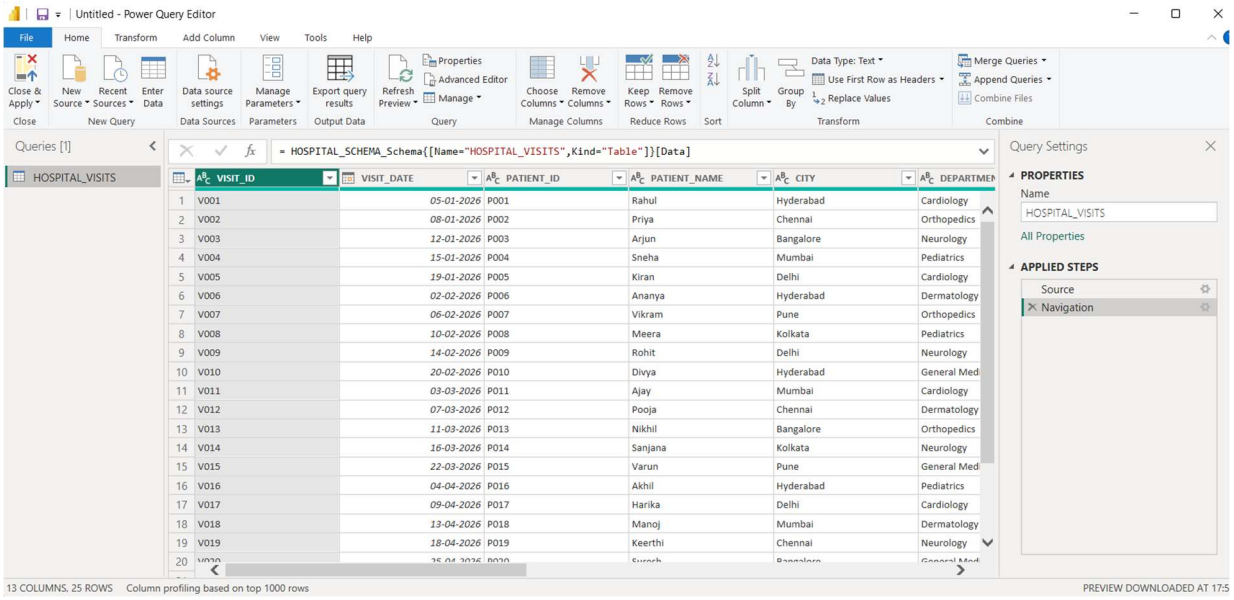




Select the Database and Click on Transform



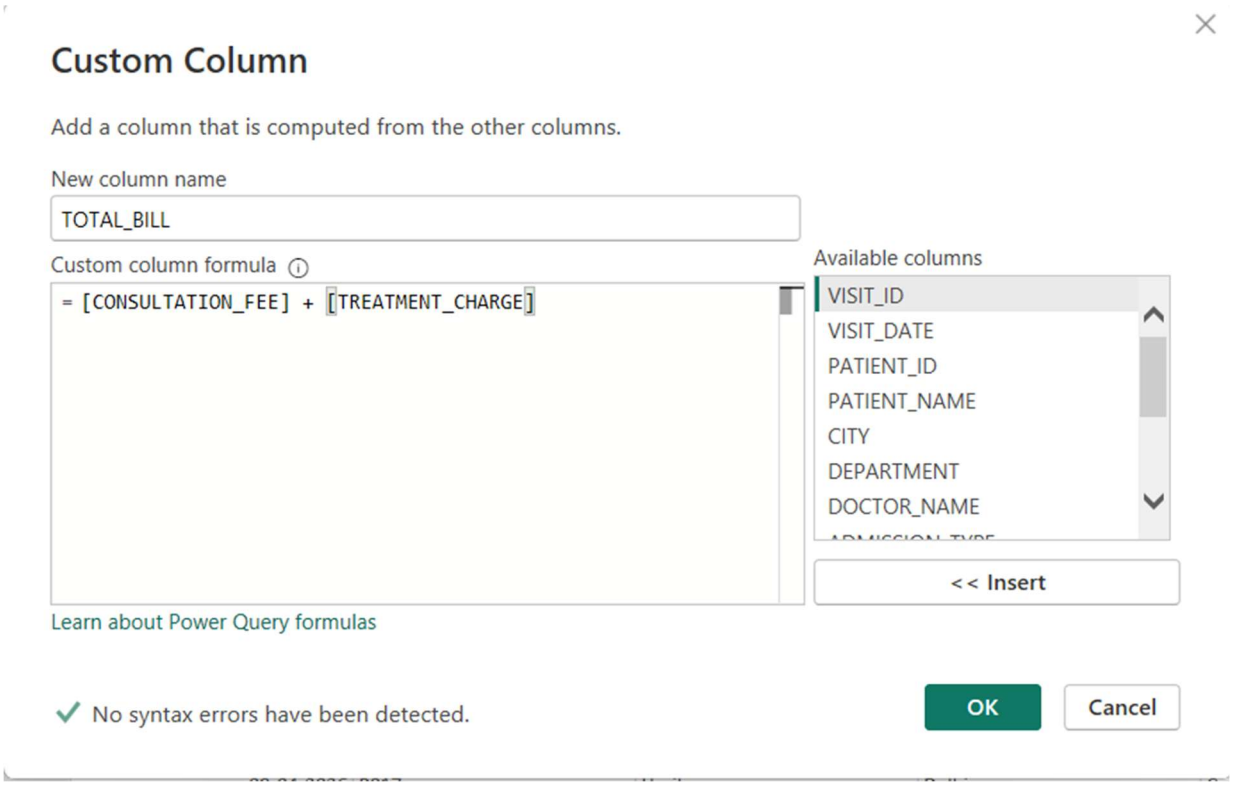
Load the Data to Power Query Editor

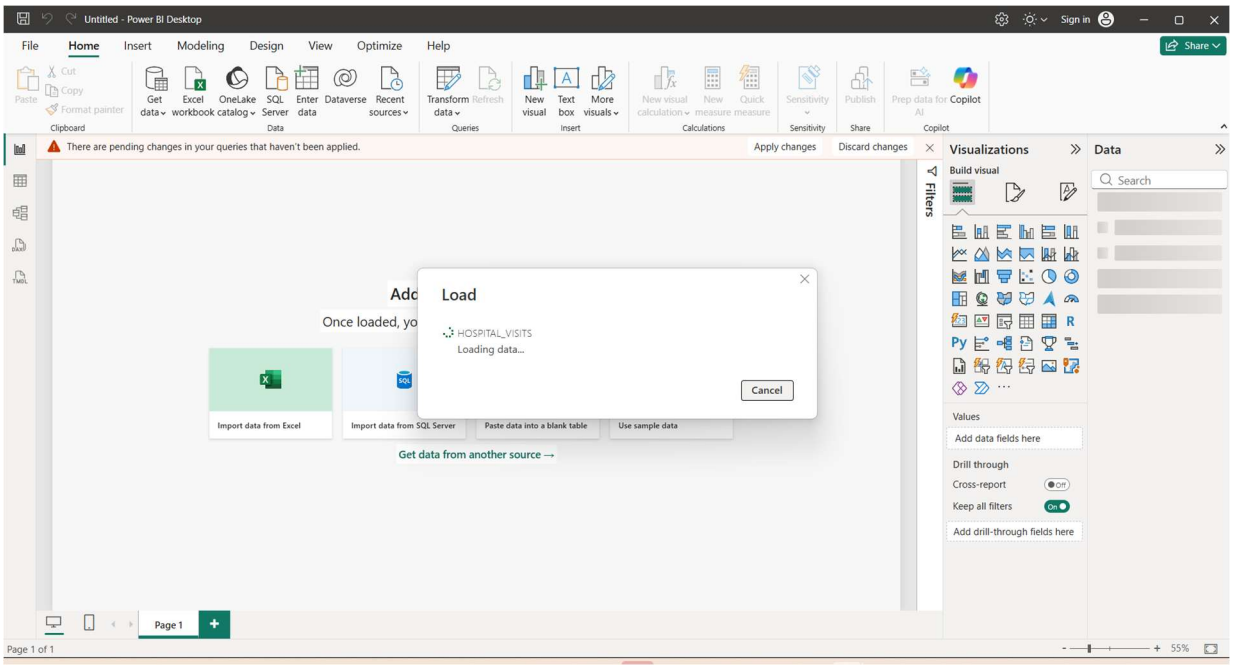


Create a new column

create a new column called TOTAL_BILL.

TOTAL_BILL = [CONSULTATION_FEE] + [TREATMENT_CHARGE]



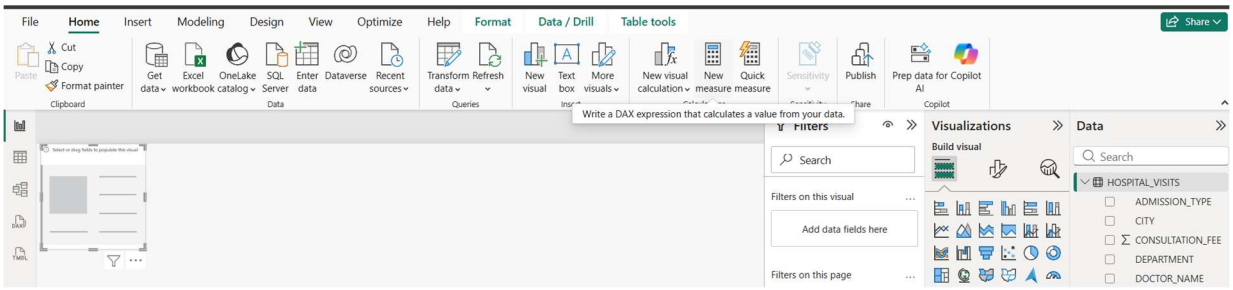


Visualizations:

Create four KPI cards using DAX:

We'll create four measures.

1. Total Patients



Click On the measure



New measure will be added

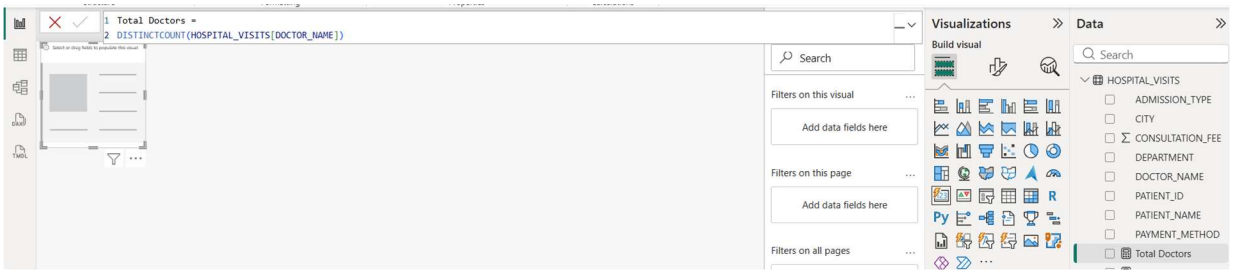
. Total Visits



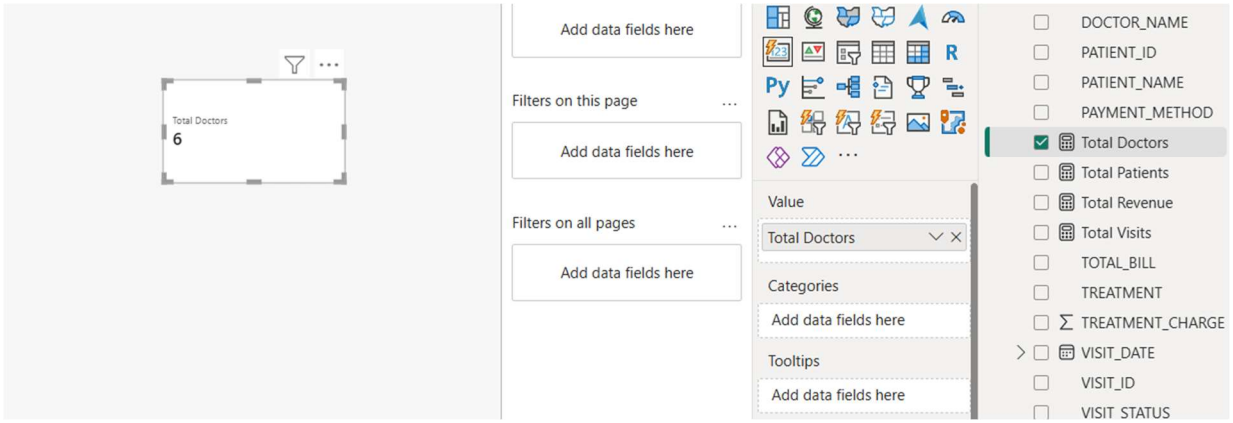
Total Revenue



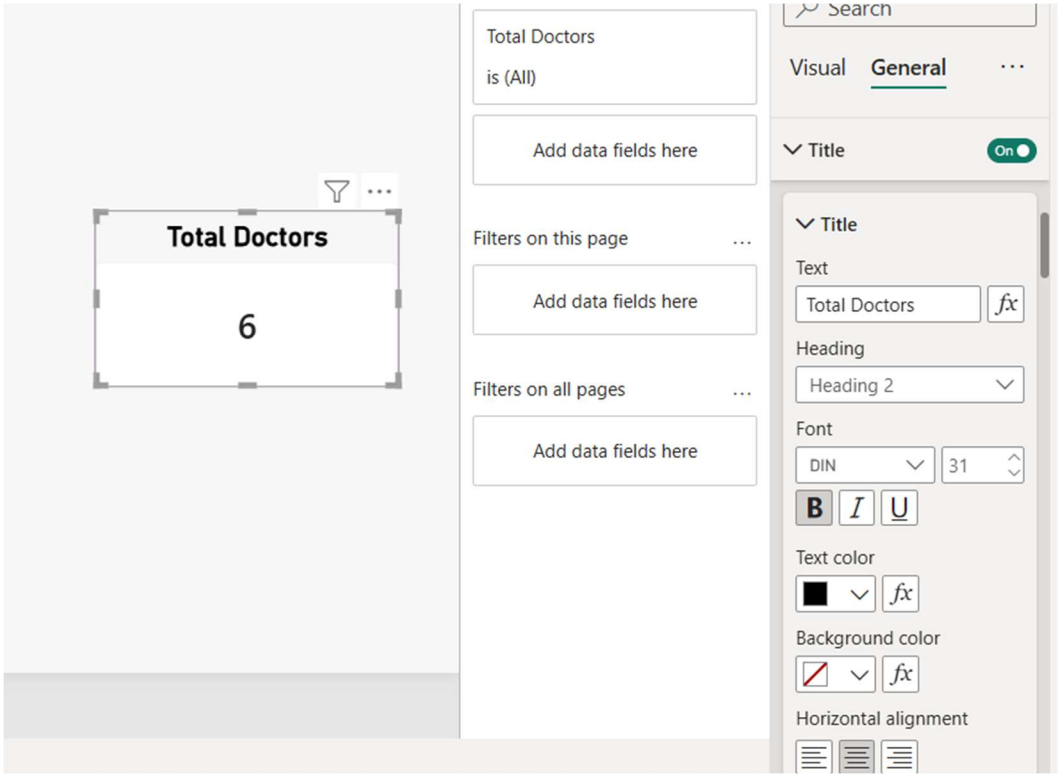
Total Doctors



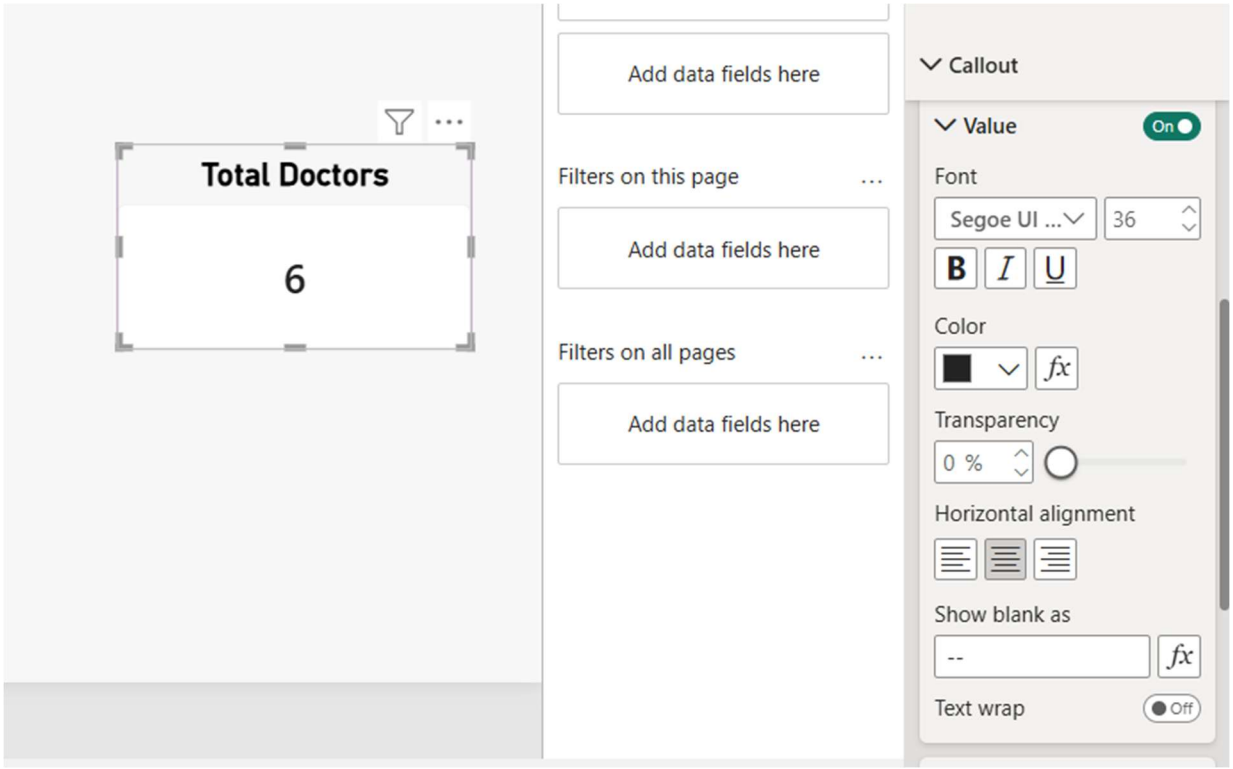
Displaying the Total Doctors KPI Card :



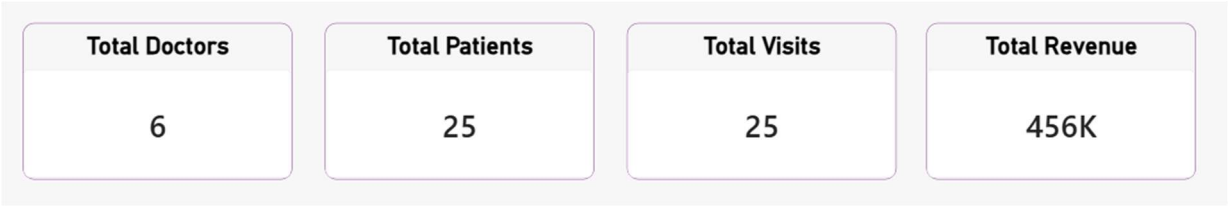
Turning on the title :



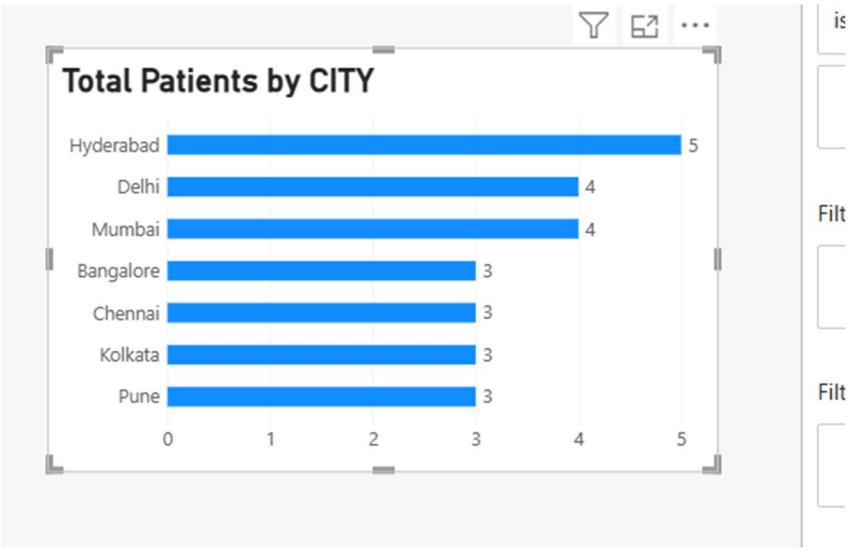
Tuning the label off and Then increasing the value of the data in the card :



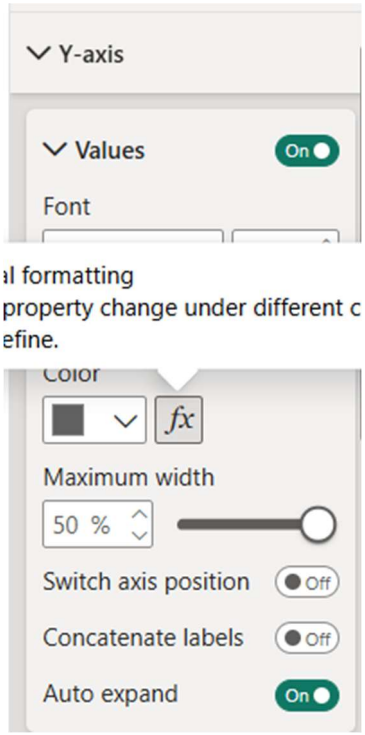
1.Similarly Adding the 4 Other Cards :

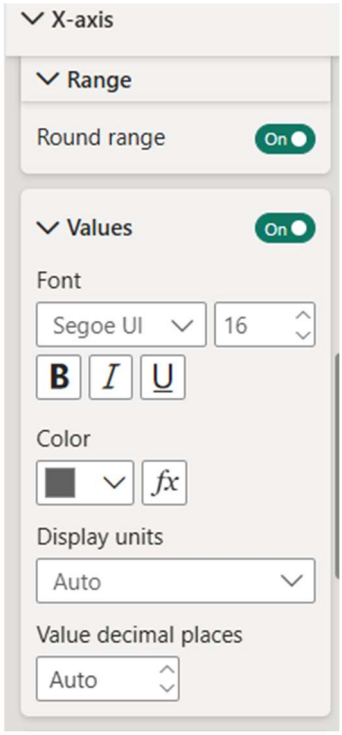


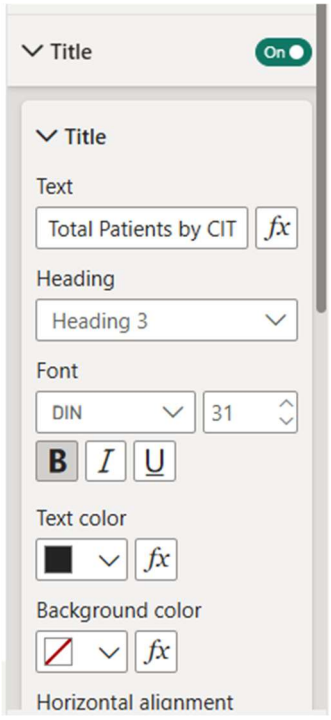
2.Patient Distribution by City(Bar Chart):



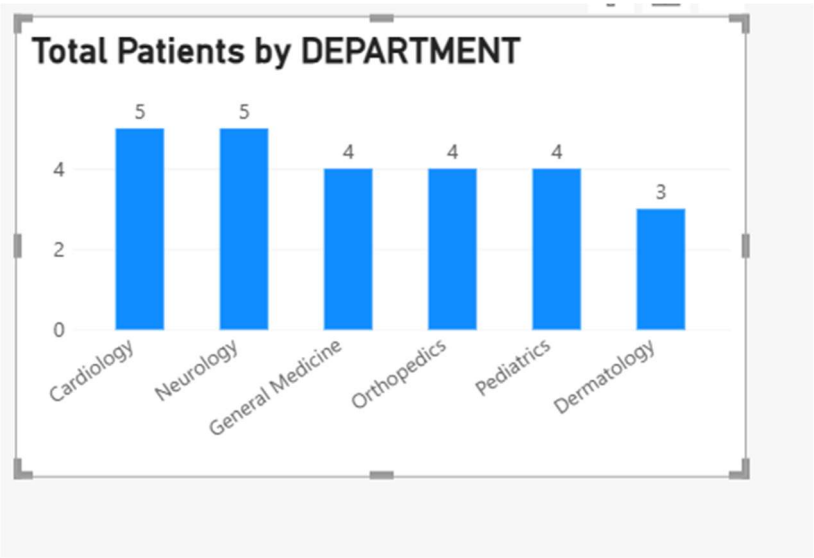
Applied Styles :



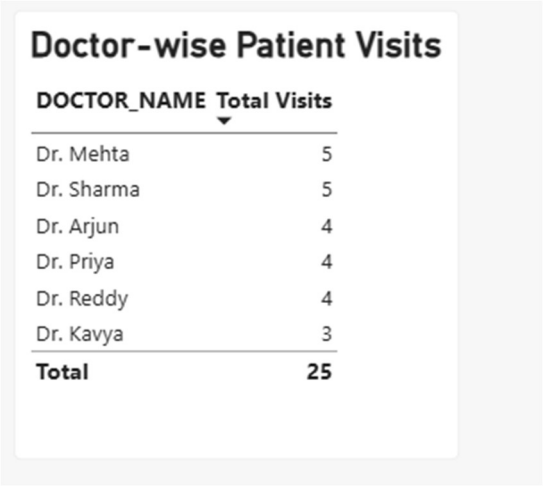




3. Department-wise Patient Count(Column chart)

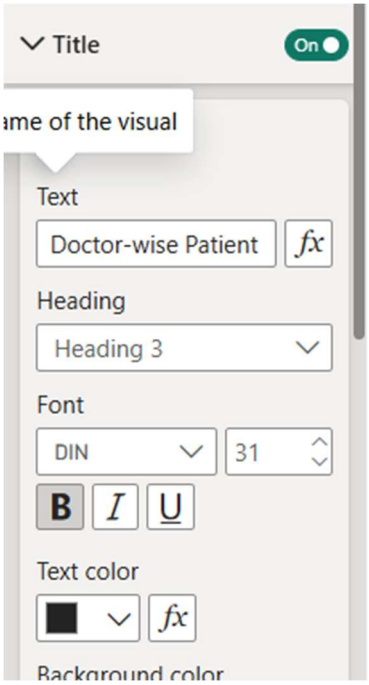


Doctor-wise Patient Visits (table)



Doctor-wise Patient Visits

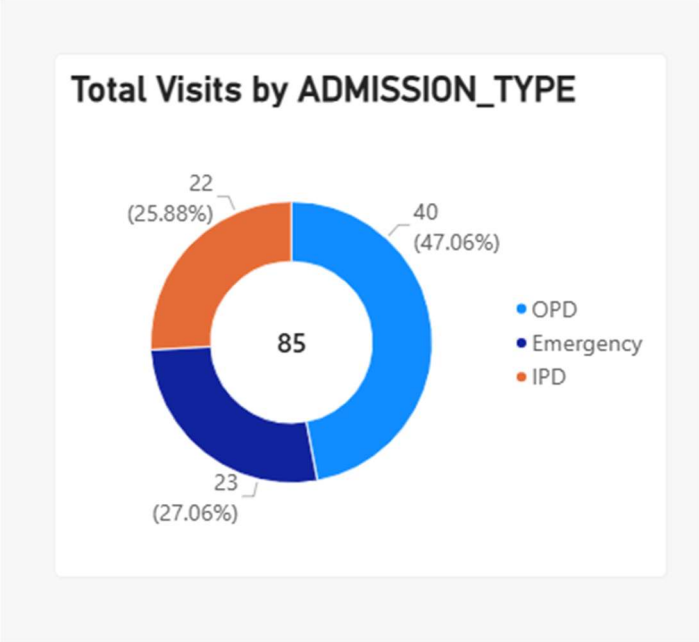
DOCTOR_NAME	Total Visits
Dr. Mehta	5
Dr. Sharma	5
Dr. Arjun	4
Dr. Priya	4
Dr. Reddy	4
Dr. Kavya	3
Total	**25**



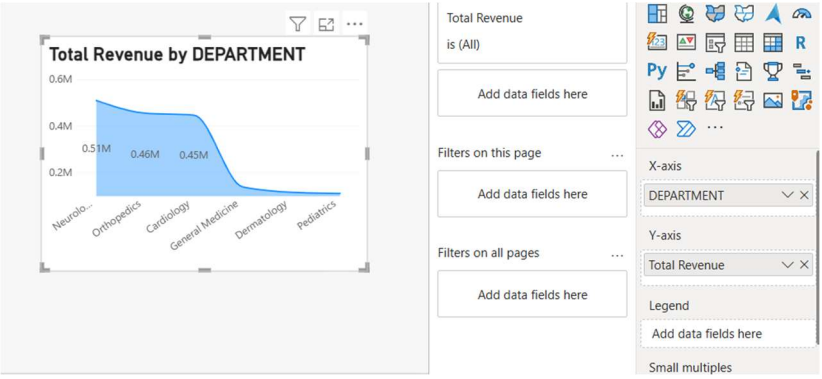
Monthly Patient Visit Trends(Line Chart)



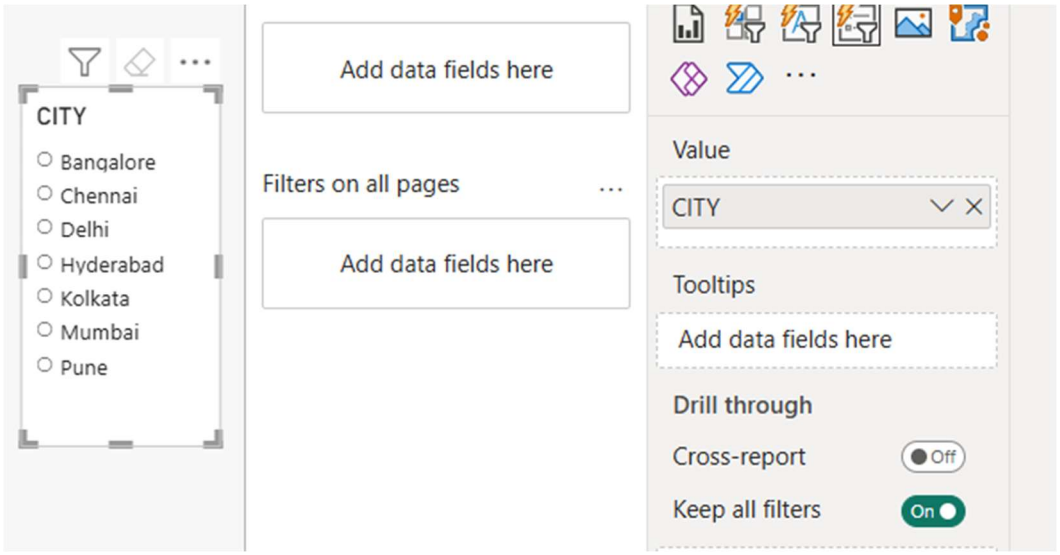
Admission Type Distribution(Donut Chart)



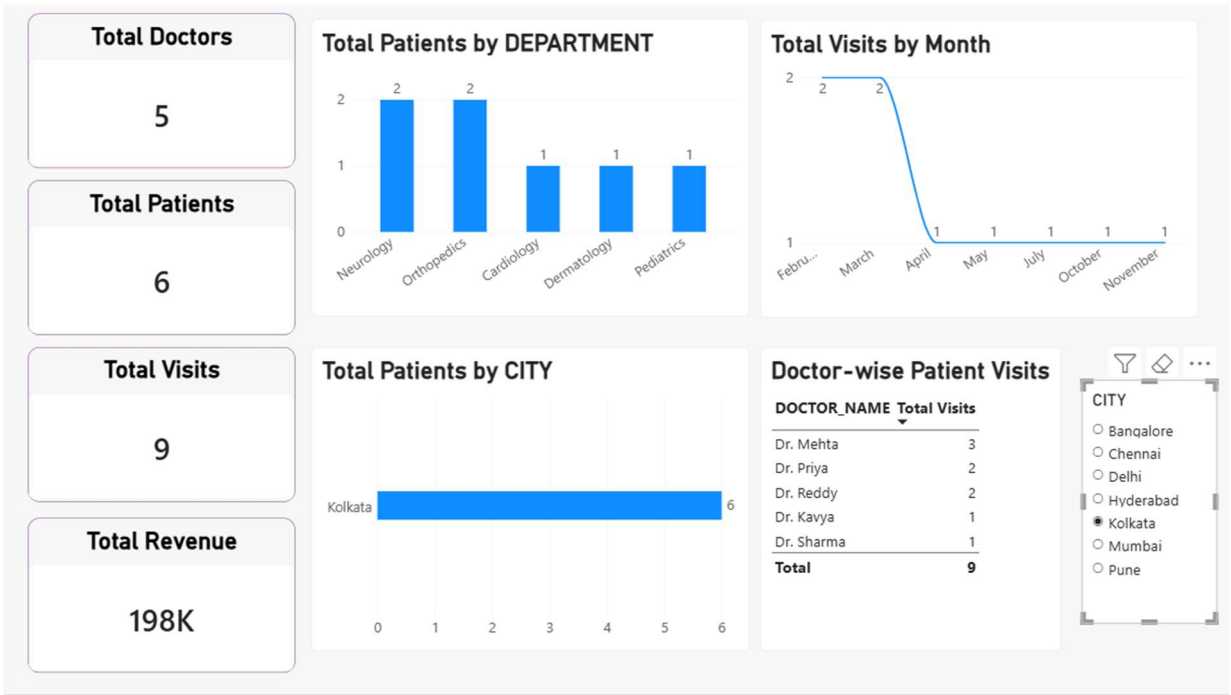
Revenue by Department(Stacked area chart)

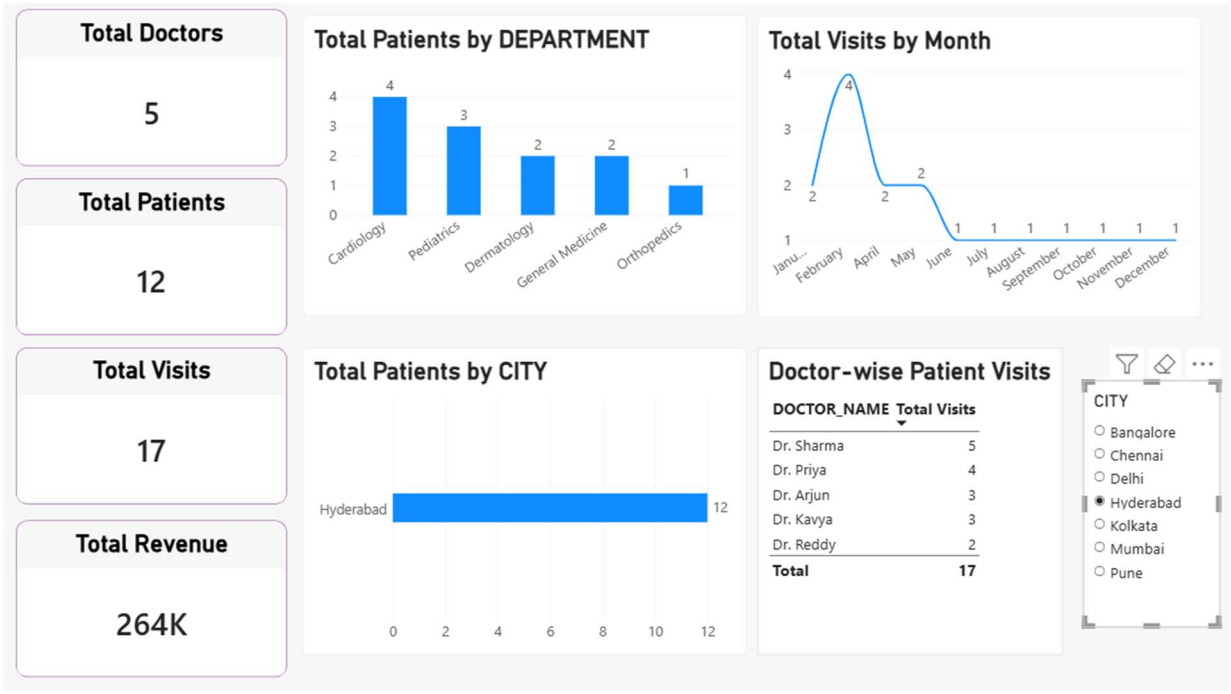


Adding the List Slicer:

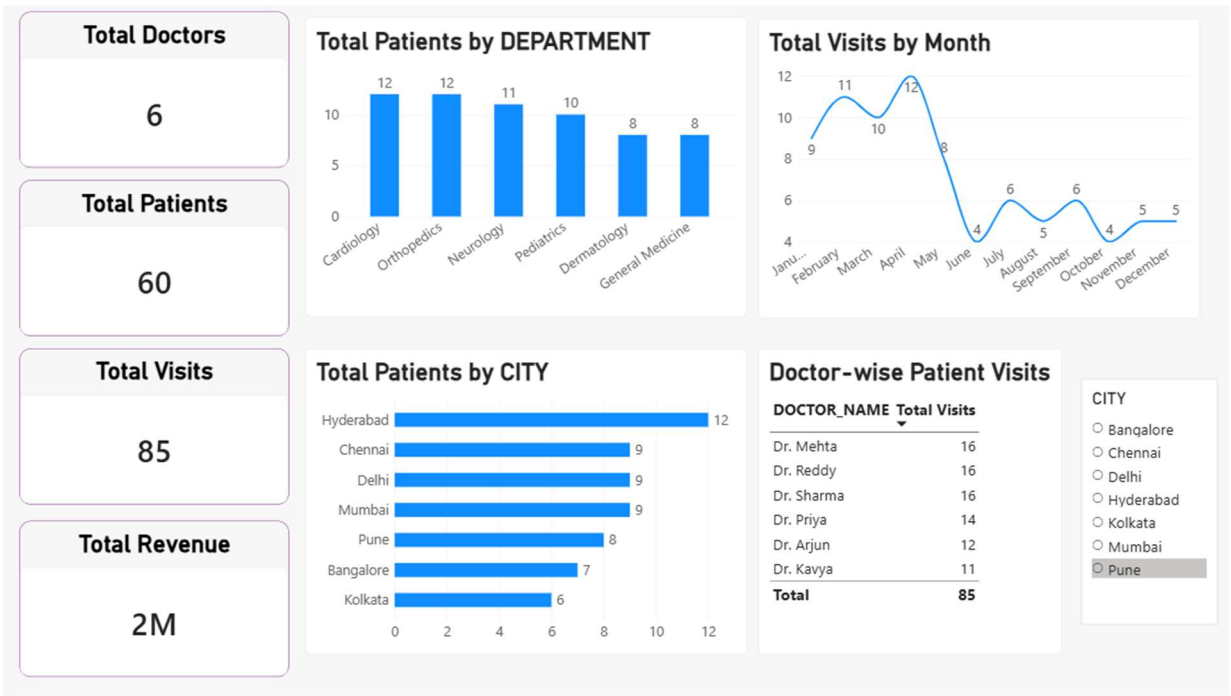


Checking the Data:

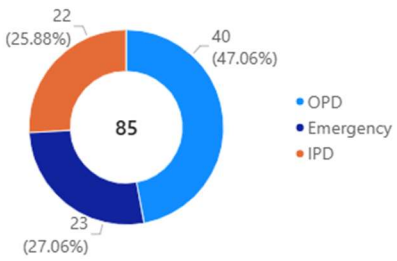




Final Dashboard:



Total Visits by ADMISSION_TYPE



Total Revenue by DEPARTMENT

